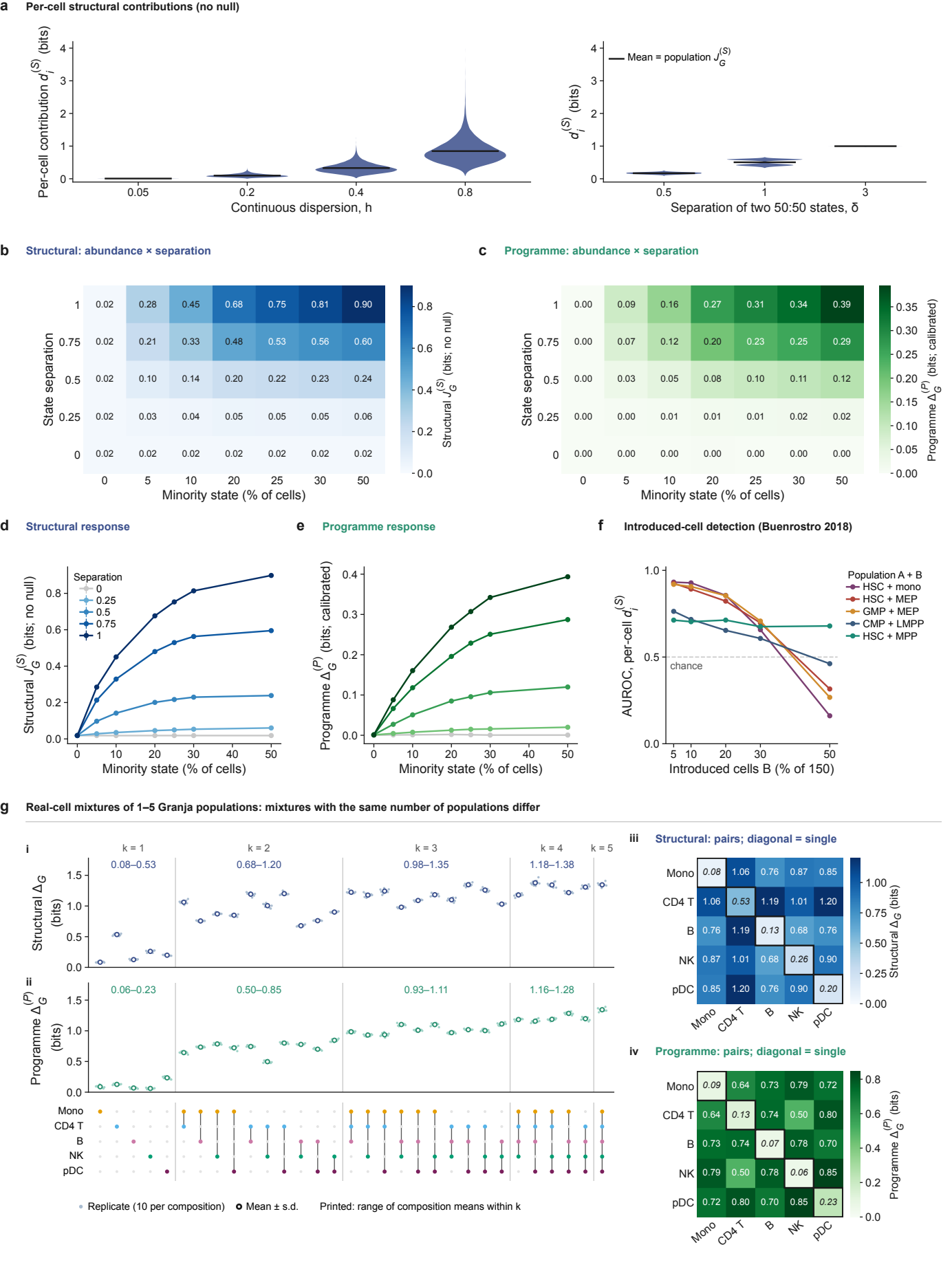

a
Per-cell structural contributions (no null)
Per-cell contribution $d_i^{(S)}$ (bits)
Continuous dispersion, h
0.05
0.2
0.4
0.8
Mean = population $J_G^{(S)}$
$d_i^{(S)}$ (bits)
Separation of two 50:50 states, δ
0.5
1
3
b
Structural: abundance × separation
State separation
Minority state (% of cells)
Structural $J_G^{(S)}$ (bits; no null)
c
Programme: abundance × separation
Programme $\Delta_G^{(P)}$ (bits; calibrated)
d
Structural response
Separation
0
0.25
0.5
0.75
1
e
Programme response
f
Introduced-cell detection (Buenrostro 2018)
AUROC, per-cell $d_i^{(S)}$
chance
Introduced cells B (% of 150)
Population A + B
HSC + mono
HSC + MEP
GMP + MEP
CMP + LMPP
HSC + MPP
g
Real-cell mixtures of 1–5 Granja populations: mixtures with the same number of populations differ
i
ii
k = 1
k = 2
k = 3
k = 4
k = 5
0.08–0.53
0.68–1.20
0.98–1.35
1.18–1.38
0.06–0.23
0.50–0.85
0.93–1.11
1.16–1.28
Structural Δ_G (bits)
Programme $\Delta_G^{(P)}$ (bits)
Mono
CD4 T
B
NK
pDC
Replicate (10 per composition)
Mean ± s.d.
Printed: range of composition means within k
iii
Structural: pairs; diagonal = single
iv
Programme: pairs; diagonal = single